

Project Planning & Milestones

Project Planning Sheet - Rising Waters

1. Project Milestone Schedule

gantt	
title Rising Waters Development Timeline	
dateFormat YYYY-MM-DD	
section Data & Model Phase	
Data Collection & Cleaning	:active, 2026-06-01, 7d
Exploratory Data Analysis (EDA)	: 2026-06-08, 5d
Model Training & Evaluations	: 2026-06-13, 6d
section Web App Development	
Form Design & Bootstrap 5 Theme	: 2026-06-19, 4d
Flask Integration & Routing	: 2026-06-23, 4d
Validation Engine & JS overlays	: 2026-06-27, 3d
section Testing & Deployment	
Unit & Integration Testing	: 2026-06-30, 2d
GitHub Setup & Vercel Upload	: 2026-07-02, 2d

2. Activity Breakdown

1. Sprint 1: Data Preparation
- Import meteorology records.
 - Clean empty values and capping outliers using Interquartile Range (IQR).
 - Visualize data correlations with heatmaps in Jupyter Notebook.
2. Sprint 2: Model Engineering & Scaling
- Partition data into 75% training and 25% testing.
 - Evaluate Decision Tree, Random Forest, KNN, and XGBoost classifiers.
 - Optimize the scaler and export parameters.
3. Sprint 3: Web Server Implementation
- Create Flask app structure with `/Predict`, `/chance`, `/no_chance` routes.
 - Structure templates with Bootstrap 5.
 - Set up client-side form validations.
4. Sprint 4: Verification & Cloud Launch
- Run local unit tests verifying redirects and validators.
 - Refactor codebase to fit Vercel bundle restrictions.
 - Initialize git, commit changes, and launch on Vercel.